Sharp Two-Contrast Gardner Bounds and a Trace-Norm Defect

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Abstract

We study two extremal problems with prescribed spectral contrasts. For $W > 0$ and $\varphi_W(X) = \mathrm{Tr}(WX)$, the sharp dimension-independent bound in Gardner's inequality is $|\varphi_W(X)| \le (1-\delta^2\delta'^2)^{-1/2}\varphi_W(|X|)$, where $\delta = \Delta(W)$ and $\delta' = \Delta(|X|)$. It is attained in even dimension; in odd dimension we give the exact constant in radicals, including its boundary cases and monotonicity in dimension. For self-adjoint X and positive Y, we also determine the exact maximum of the trace-norm defect $\mathrm{Tr}(Y|X|Y) - \mathrm{Tr}\,|YXY|$ on direct sums of matrix algebras with prescribed contrasts and operator norms. Its parity correction requires a matrix linearisation when an extreme eigenvector is not confined to one principal-angle block. Both results have analytic proofs. The classical Kantorovich inequality, trace-norm duality and two-projection decomposition are stated as tools, separately from the quantitative extremal formulas.

Keywords: Gardner inequality; positive functional; spectral contrast; trace norm; Kantorovich inequality; two projections.
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1 Introduction

Gardner's inequality $|\varphi(X)| \le \varphi(|X|)$ characterises traces among positive functionals on a C^*-algebra [1, 2]. On $M_n(\mathbb{C})$ write $\varphi_W(X) = \mathrm{Tr}(WX)$, with $W > 0$. The functional is tracial precisely when W is scalar. Its departure from a scalar matrix can be measured by the spectral, or Michelson, contrast

$$\Delta(W) = \frac{\lambda_{\max}(W) - \lambda_{\min}(W)}{\lambda_{\max}(W) + \lambda_{\min}(W)}.$$

Abed, Nikolaeva and Novikov [3] studied operator contrast and sampled trace functionals numerically. Here we give analytic extremal formulas for two related problems with prescribed spectral contrasts.

The first problem asks for the largest Gardner quotient when the contrasts of both W and $|X|$ are fixed. The unrestricted-modulus estimate is an application of the classical Kantorovich inequality [4]. The joint constraint gives the sharper constant $(1 - \Delta(W)^2\Delta(|X|)^2)^{-1/2}$. It is attained in every even dimension. An unavoidable scalar summand lowers the odd-dimensional optimum when both contrasts are nonzero and $|X|$ is invertible. We evaluate this optimum explicitly and prove its monotonicity with dimension.

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The second problem concerns

$$\mathcal{D}(X,Y) = \operatorname{Tr}(Y|X|Y) - \operatorname{Tr}|YXY|, \qquad X = X^*, \quad Y \geq 0.$$

This nonnegative defect of the trace-norm triangle inequality depends on the interaction of the positive and negative spectral subspaces of X after congruence by Y. With operator norms one and both contrasts prescribed, we determine its maximum on $\bigoplus_c M_{k_c}$. The number $\sum_c \lfloor k_c/2 \rfloor$ counts the available two-dimensional subspaces, while the presence of an odd block changes whether the smallest eigenvalue of $|X|$ reduces the maximum. A separate matrix linearisation handles an extreme eigenvector spread across several such subspaces; simple blockwise optimisation alone does not cover it.

The common tools are trace-norm duality [6], spectral endpoint reduction and the two-projection decomposition [5]. The characterisation, Kantorovich bound and decomposition are classical. The results below specify the joint-contrast extrema and their parity effects within these classical reduction tools.

This focused article develops material from the authors' combined manuscript *How characterisations of the trace fail: sharp violation bounds in terms of Michelson contrasts* [7]. The joint-contrast Gardner and trace-norm-defect formulas were included there. The present account isolates these two problems with full proofs and supplements that account with the explicit odd-dimensional evaluation. Other unnormalised functionals, projection-lattice tests, commutator thresholds and general measurable-density extensions are outside this focused paper. Sections 2–3 give the Gardner proof; Section 4 proves the trace-norm-defect formula in full.

2 Notation and reduction tools

We write $\|A\|$ and $\|A\|_1 = \operatorname{Tr}|A|$ for the operator and trace norms, with $|A| = (A^*A)^{1/2}$. All matrix traces are unnormalised. On $\mathcal{M} = \bigoplus_{c=1}^{C} M_{k_c}(\mathbb{K})$, $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, use $\operatorname{Tr} = \sum_c \operatorname{Tr}_c$; "unitary" means orthogonal in the real case. Contrast is assigned only to nonzero positive matrices; a singular such matrix has contrast one. Put $\kappa(\delta) = (1-\delta^2)^{-1/2}$ for $0 \leq \delta < 1$.

Lemma 2.1 (Classical Gardner bound)**.** *For $W > 0$ with extreme eigenvalues $a \geq b > 0$, every matrix X satisfies*

$$|\operatorname{Tr}(WX)| \leq \frac{a+b}{2\sqrt{ab}} \operatorname{Tr}(W|X|).$$

The constant is attained for the fixed density W.

Proof. Write $X = UT$, $T = |X|$, extending the polar factor to a unitary. Hilbert–Schmidt Cauchy–Schwarz gives $|\operatorname{Tr}(WUT)|^2 \leq \operatorname{Tr}(W^2T)\operatorname{Tr}T$. The positive functional $\omega(B) = \operatorname{Tr}(TB)$ satisfies $\omega(W^2) \leq (a+b)\omega(W) - ab\omega(I)$, by $(aI - W)(W - bI) \geq 0$. Writing $x = \omega(W)$, $y = \omega(I)$, the identity

$$\frac{(a+b)^2}{4ab}x^2 - ((a+b)x - aby)y = \left(\frac{a+b}{2\sqrt{ab}}x - \sqrt{ab}\,y\right)^2$$

proves the Kantorovich estimate and hence the claim. For attainment when $a > b$, take eigenvectors e_a, e_b of W for a, b, put $v = \sqrt{b/(a+b)}e_a + \sqrt{a/(a+b)}e_b$, and choose a unitary with $Uv = Wv/\|Wv\|$. The matrix $X = Uvv^*$ gives equality. If $a = b$, any nonzero positive X gives equality. □

Lemma 2.2 (Spectral endpoint principle)**.** *Fix $0 \le \ell < 1$ and a matrix eigenbasis. A continuous convex function of the eigenvalues on $[\ell, 1]^n$ attains its maximum at a vertex. The same is true on any face obtained by prescribing one eigenvalue to be ℓ and another to be 1. For $c \in \mathbb{R}$ the function*

$$(W, T) \longmapsto \|TW\|_1 - c \operatorname{Tr}(TW)$$

is separately convex in the eigenvalues of W and T when their eigenbases are fixed.

Proof. Every point of a cube or one of its faces is a convex combination of its vertices. Convexity bounds its value by the largest vertex value. For the last statement, multiplication by a fixed matrix is linear, the trace norm is convex, and the trace term is linear. Holding one matrix fixed gives the asserted convexity for the other. □

Lemma 2.3 (Two projections)**.** *Two orthogonal projections on a finite-dimensional real or complex Hilbert space admit an orthogonal decomposition into common invariant subspaces of dimension one or two. On a noncommuting two-dimensional summand both projections have rank one.*

Proof. On $\operatorname{ran} P$ diagonalise the self-adjoint contraction PQP. For an eigenvector v with eigenvalue $c \in (0, 1)$ put $w = (Qv - cv)/\sqrt{c(1-c)}$. Then $w \in \ker P$, $\|w\| = 1$, and

$$Qv = cv + \sqrt{c(1-c)}\, w, \qquad Qw = \sqrt{c(1-c)}\, v + (1-c)w.$$

Thus $\operatorname{span}\{v, w\}$ is invariant under both projections. Orthogonal eigenvectors, including within a repeated eigenspace, give orthogonal pairs. The eigenvalues zero and one of PQP give common eigenvectors of P, Q; on the remaining subspace of $\ker P$, diagonalise Q. This completes the decomposition. It is the finite-dimensional form of the two-subspace theorem [5]. □

Lemma 2.4. *Let $W, T \ge 0$ be two-by-two matrices with eigenvalues $x_1 \ge x_2 \ge 0$ and $y_1 \ge y_2 \ge 0$. Let $\gamma \in [0, 1]$ be the squared overlap of unit eigenvectors for x_1, y_1, and set*

$$S = x_1 y_2 + x_2 y_1, \quad q = (x_1 + x_2)(y_1 + y_2), \quad p = (x_1 - x_2)(y_1 - y_2).$$

Then

$$\operatorname{Tr}(TW) = S + \gamma p, \qquad \|TW\|_1^2 = S^2 + \gamma p q. \tag{1}$$

Proof. Expanding the trace in the two eigenbases gives the first identity. If s_1, s_2 are the singular values of TW, then $s_1^2 + s_2^2 = \operatorname{Tr}(W^2 T^2)$ and $s_1 s_2 = x_1 x_2 y_1 y_2$. Expanding the former expression in the same eigenbases and adding twice the latter gives the second identity. Every $\gamma \in [0, 1]$ is realised by a real rotation. □

3 The exact two-contrast Gardner constant

For $0 \le \delta < 1$, $0 \le \delta' \le 1$ define

$$R_n(\delta, \delta') = \sup \left\{ \frac{|\operatorname{Tr}(WX)|}{\operatorname{Tr}(W|X|)} : W > 0,\ X \ne 0,\ \Delta(W) = \delta,\ \Delta(|X|) = \delta' \right\}.$$

The denominator is positive because $W > 0$ and $X \ne 0$.

Theorem 3.1. *For every $n \geq 2$,*

$$R_n(\delta, \delta') \leq C(\delta, \delta') := \frac{1}{\sqrt{1 - \delta^2 \delta'^2}}. \tag{2}$$

If n is even, equality holds. For odd $n = 2k + 1$, set

$$\ell = \frac{1 - \delta}{1 + \delta}, \quad \ell' = \frac{1 - \delta'}{1 + \delta'}, \quad S = \ell + \ell', \quad q = (1 + \ell)(1 + \ell'), \quad p = (1 - \ell)(1 - \ell').$$

Then

$$R_{2k+1}(\delta, \delta') = \max_{0 \leq \gamma \leq 1} \frac{k\sqrt{S^2 + \gamma pq} + \ell\ell'}{k(S + \gamma p) + \ell\ell'}. \tag{3}$$

The maxima are attained over real matrices as well as complex matrices. If $0 < \delta < 1$ and $0 < \delta' < 1$, the odd-dimensional value is strictly smaller than $C(\delta, \delta')$ and converges to it as $k \to \infty$. If $\delta' = 1$, it equals $\kappa(\delta)$ in every dimension. If either contrast is zero, it equals one.

Proof. Write $X = UT$, $T = |X|$. Trace-norm duality gives

$$\sup_U |\operatorname{Tr}(WUT)| = \|TW\|_1,$$

with a unitary maximiser obtained from a polar decomposition. Rescale W, T so that their largest eigenvalues are one. To prove a proposed bound $r \geq 1$ it suffices to prove $\|TW\|_1 - r \operatorname{Tr}(TW) \leq 0$. Lemma 2.2, applied successively on the faces fixing the smallest and largest eigenvalues, reduces this verification to $W = \ell I + (1 - \ell)P$, $T = \ell' I + (1 - \ell')Q$. The cases of a zero contrast are immediate, so assume both positive. By Lemma 2.3, the matrices split into one- and two-dimensional blocks. A one-dimensional block contributes the same amount to numerator and denominator. On a two-dimensional block Lemma 2.4 gives contributions

$$s(\gamma) = \sqrt{S^2 + \gamma pq}, \qquad t(\gamma) = S + \gamma p.$$

Differentiation shows that $s(\gamma)^2/t(\gamma)^2$ is maximal at $\gamma = S/q$, with value

$$\frac{q^2}{4S(S + p)} = \frac{1}{1 - \delta^2 \delta'^2}.$$

Here $S > 0$ and $0 < S/q \leq 1$. The ratio of sums of nonnegative contributions cannot exceed the largest individual ratio. This proves (2). If n is even, use $n/2$ copies of this block and the polar maximising unitary.

For completeness we verify the exact parity correction. Put $\mu = \ell\ell'$. A one-dimensional block contributes one of $1, \ell, \ell', \mu$, hence at least μ. For a test value $1 \leq r \leq C$, its contribution to the additive score $\|TW\|_1 - r \operatorname{Tr}(TW)$ is at most $\mu(1 - r)$. A two-dimensional block contributes at most

$$f_r = \max_{0 \leq \gamma \leq 1} \{s(\gamma) - rt(\gamma)\}.$$

Because a two-dimensional ratio attains C, one has $f_r \geq 0$. Replacing two scalar blocks by a two-dimensional block cannot reduce the largest possible score: it replaces $2\mu(1 - r) \leq 0$ by $f_r \geq 0$. Thus in odd dimension the largest score is achieved with k identical two-dimensional blocks and one scalar block of value μ. All these choices coexist, for example with $W = \operatorname{diag}(1, \ell)^{\oplus k} \oplus \ell$ and the corresponding T ending in ℓ'. The largest ratio is therefore exactly (3). Equivalently, at its

optimum r the maximal score is zero; the blockwise argument and the endpoint principle bound every other configuration by the same value.

When $\ell\ell' > 0$ and both contrasts are nonzero, the scalar block has positive weight and ratio one, while the two-dimensional ratio is at most $C > 1$. Hence the maximum is strictly smaller than C. Dividing numerator and denominator of (3) by k proves convergence to C. For $\ell' = 0$ the extra scalar block vanishes, so the two-dimensional maximum is attained without dilution. □

Corollary 3.2 (Closed form and monotonicity in odd dimension)**.** *Let $n = 2k + 1$ and use the notation of Theorem 3.1. Put $t = \ell\ell'$ and*

$$y_* = \frac{\sqrt{t^2 + k^2 S(q - S) + kqt} - t}{k}. \tag{4}$$

Then

$$R_{2k+1}(\delta, \delta') = \frac{q}{2y_*}. \tag{5}$$

If $p > 0$, the unique maximising overlap in (3) *is $\gamma_* = (y_*^2 - S^2)/(pq)$ and belongs to $(0, 1)$. If $p = 0$, formula* (5) *equals one and every overlap is a maximiser. For $0 < \delta, \delta' < 1$ the constants R_{2k+1} increase strictly with k to $C(\delta, \delta')$. If $\delta' = 1$, they all equal $C(\delta, 1)$.*

Proof. Assume first $p > 0$. Since $q = 2S + p$, the substitution $y = \sqrt{S^2 + \gamma pq}$ maps $[0, 1]$ onto $[S, q - S]$. The quotient in (3) becomes

$$r_k(y) = \frac{q(ky + t)}{ky^2 + kS(q - S) + qt}.$$

Its derivative has the opposite sign to $H(y) = ky^2 + 2ty - kS(q - S) - qt$. On the positive interval this polynomial is strictly increasing, and

$$H(S) = -p(kS + t) < 0, \qquad H(q - S) = p\big(k(q - S) + t\big) > 0.$$

There is exactly one critical point, it is the global maximum, and solving $H(y_*) = 0$ gives (4). The same equation rewrites the denominator of $r_k(y_*)$ as $2y_*(ky_* + t)$, proving (5). If $p = 0$, then $q = 2S$ and the square root in (4) is $t + kS$, so the formula gives one. Thus it also covers zero contrasts without a limiting argument. When $t = 0$, it gives $y_* = \sqrt{S(q - S)}$ and the uniform constant.

For strict monotonicity suppose $p, t > 0$. Write $y_0 = \sqrt{S(q - S)}$. One has $H(y_0) = t(2y_0 - q) < 0$ and $H(q/2) = kp^2/4 > 0$, so $y_0 < y_* < q/2$. Treating k as a positive real variable and differentiating its defining equation gives

$$\frac{dy_*}{dk} = -\frac{y_*^2 - S(q - S)}{2ky_* + 2t} < 0.$$

Consequently $q/(2y_*)$ increases strictly. Equation (4) yields $y_* \to \sqrt{S(q - S)}$, so the limit is the two-dimensional constant computed in Theorem 3.1. □

Remark 3.3 (Evaluation without subtraction)*.* An equivalent expression

$$y_* = \frac{kS(q - S) + qt}{\sqrt{t^2 + k^2 S(q - S) + kqt} + t}$$

avoids subtracting close quantities in (4). For $p > 0$ an equivalent overlap formula is

$$\gamma_* = \frac{(kS + t)(y_* + S)}{q\big(k(y_* + S) + 2t\big)}.$$

Indeed $H(y_*) - H(S) = p(kS + t)$ gives $(y_* - S)\big(k(y_* + S) + 2t\big) = p(kS + t)$. These are algebraic identities, not numerical approximations to the maximiser.

4 A nonlinear trace-norm defect

Let $\mathcal{M} = \bigoplus_{c=1}^{C} M_{k_c}(\mathbb{K})$, and put

$$\nu = \sum_c \lfloor k_c/2 \rfloor.$$

For a self-adjoint X and positive Y define $\mathcal{D}(X,Y) = \operatorname{Tr}(Y|X|Y) - \operatorname{Tr}|YXY|$. In the norm-one extremal problem below write

$$\ell = \frac{1-\delta}{1+\delta}, \qquad \ell' = \frac{1-\delta'}{1+\delta'}, \qquad \delta = \Delta(|X|) < 1, \quad \delta' = \Delta(Y) \le 1.$$

The first observation identifies when the defect vanishes.

Proposition 4.1. *For $X = X^*$ and $Y \ge 0$, $\operatorname{Tr}|YXY| \le \operatorname{Tr}(Y|X|Y)$, with equality if and only if $YX_+Y^2X_-Y = 0$, in particular if Y commutes with X.*

Proof. With $X = X_+ - X_-$ and $|X| = X_+ + X_-$, the triangle inequality for the trace norm gives $\|YX_+Y - YX_-Y\|_1 \le \operatorname{Tr}(YX_+Y) + \operatorname{Tr}(YX_-Y)$. For positive A, B, $\|A-B\|_1 = \operatorname{Tr}A + \operatorname{Tr}B$ if and only if $AB = 0$: with P the projection onto the positive part of $A-B$, $\|A-B\|_1 = \operatorname{Tr}A + \operatorname{Tr}B - 2\operatorname{Tr}((1-P)A) - 2\operatorname{Tr}(PB)$, and equality forces $(1-P)A = 0 = PB$, so $AB = AP(1-P)B = 0$; conversely, if $AB = 0$ the supports are orthogonal and $|A-B| = A+B$. Apply this to $A = YX_+Y$, $B = YX_-Y$. If $XY = YX$, then $YX_+Y^2X_-Y = Y^4X_+X_- = 0$. □

Proposition 4.2. *Let $X = X^*$ in $M_2(\mathbb{C})$, let $|X|$ have eigenvalues $x_1 \ge x_2 \ge 0$, $x_1 > 0$, and let $Y \ge 0$ have eigenvalues $y_1 \ge y_2 \ge 0$. Then*

$$\operatorname{Tr}(Y|X|Y) - \operatorname{Tr}|YXY| \le \frac{2x_1x_2}{x_1+x_2}\,(y_1-y_2)^2.$$

With $S = x_1y_2 + x_2y_1$, $q = (x_1+x_2)(y_1+y_2)$ and $\gamma = |\langle e_1, f_1\rangle|^2$ for unit eigenvectors e_1 of X for its eigenvalue of modulus x_1 and f_1 of Y for y_1: if $x_2 > 0$ and $y_1 > y_2$, the bound is attained if and only if X is indefinite and $\gamma = S/q$; it is attained by real matrices.

Proof. $\mathcal{D}$ is invariant under $X \mapsto -X$ and vanishes for semidefinite X. Otherwise let $Xe_1 = x_1e_1$, $Xe_2 = -x_2e_2$, $u = \langle e_1, Y^2e_1\rangle = \gamma y_1^2 + (1-\gamma)y_2^2$ and $s = y_1^2 + y_2^2$. Then $\operatorname{Tr}(Y|X|Y) = \operatorname{Tr}(|X|Y^2) = A := (x_1-x_2)u + x_2s$, and since $\det(YXY) = -x_1x_2y_1^2y_2^2 \le 0$, $\operatorname{Tr}|YXY| = \sqrt{\operatorname{Tr}(YXY)^2 - 4\det(YXY)} = \sqrt{L^2+K^2}$ with $L = \operatorname{Tr}(XY^2) = (x_1+x_2)u - x_2s$ and $K = 2\sqrt{x_1x_2}\,y_1y_2$. For the unit vector $(\alpha,\beta) = (x_1-x_2, 2\sqrt{x_1x_2})/(x_1+x_2)$ the Cauchy–Schwarz inequality gives $\sqrt{L^2+K^2} \ge \alpha L + \beta K$, and $A - \alpha L - \beta K = \frac{2x_1x_2}{x_1+x_2}(y_1-y_2)^2$ for every u; equality holds iff $\beta L = \alpha K$, i.e. $u - y_2^2 = (y_1-y_2)S/(x_1+x_2)$, i.e. $\gamma = S/q$. Real maximisers: diagonal X and Y rotated by $\arccos\sqrt{\gamma}$. □

Theorem 4.3. *Let $\mathcal{M} = \bigoplus_{c=1}^{C} M_{k_c}(\mathbb{K})$, $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, with $\nu \ge 1$, and let $X = X^*$ and $Y \ge 0$, with $|X|$ and Y normalised to $\lambda_{\max} = 1$, with $\Delta(|X|) = \delta \in [0,1)$ and $\Delta(Y) = \delta' \in [0,1]$. Then*

$$\sup\bigl(\operatorname{Tr}(Y|X|Y) - \operatorname{Tr}|YXY|\bigr) = \begin{cases} (\nu-\delta)\bigl(\frac{2\delta'}{1+\delta'}\bigr)^2 & \text{if every } k_c \text{ is even,} \\ \nu\bigl(\frac{2\delta'}{1+\delta'}\bigr)^2 & \text{if some } k_c \text{ is odd,} \end{cases}$$

and the supremum is attained by real direct sums of the 2×2 maximisers of Proposition 4.2 and 1×1 blocks.

For odd blocks the eigenvalue ℓ of $|X|$ can sit on a one-dimensional summand, where it costs nothing, so the supremum does not depend on δ; for $k=3$, $C=1$: $(2\delta'/(1+\delta'))^2 > (1-\delta)(2\delta'/(1+\delta'))^2$. The proof reduces to two-level spectra and, with Lemma 2.3, to 2×2 blocks, except in one case: the eigenvector of $|X|$ for ℓ may be spread over all two-dimensional blocks of its sector, so that X is not block diagonal. This case is settled by the following sharp bound, whose proof linearises the problem.

Proposition 4.4. *Let $X = X_+ \oplus (-1)$ on $\mathbb{C}^m \oplus \mathbb{C}^m$, where $0 < X_+ \le 1$ has eigenvalues $x_1, \dots, x_m$; let $c_\theta = \operatorname{diag}(\cos\theta_j)$, $s_\theta = \operatorname{diag}(\sin\theta_j)$, Q the projection onto the range of $\binom{c_\theta}{s_\theta} : \mathbb{C}^m \to \mathbb{C}^{2m}$, $b \in [0,1)$ and $Y = b + (1-b)Q$ (a two-level operator). Then*

$$\operatorname{Tr}(Y|X|Y) - \operatorname{Tr}|YXY| \;\le\; (1-b)^2 \sum_{i=1}^{m} \frac{2x_i}{1+x_i},$$

with equality for $X_+ = \operatorname{diag}(x_1,\dots,x_m)$ and $\cos^2\theta_i = (1+bx_i)/((1+x_i)(1+b))$.

Proof. Let $\gamma = 1-b^2$, $K = 1+X_+$, $A = c_\theta^2$ (so $s_\theta^2 = 1-A$). In the decomposition $\mathbb{C}^m \oplus \mathbb{C}^m$, $Y^2 = b^2 + \gamma Q$ has the blocks $b^2+\gamma A$, $\gamma c_\theta s_\theta$, $\gamma c_\theta s_\theta$, $b^2+\gamma s_\theta^2$, all diagonal. For $\lambda > 0$ the lower right block of $\lambda - XY^2$ is the invertible diagonal $E = \lambda + 1 - \gamma A$, and as $(b^2+\gamma A)E - \gamma^2 A s_\theta^2 = \lambda(b^2+\gamma A) + b^2$, the Schur complement formula gives

$$\det(\lambda - XY^2) = \det\bigl(\lambda E - X_+(\lambda(b^2+\gamma A) + b^2)\bigr) = \det\bigl((\lambda+1)(\lambda - b^2X_+) - \lambda\gamma K A\bigr) = p(\lambda),$$

$p(\lambda) = \det\bigl((\lambda+1)(\lambda - b^2X_+) - \lambda\gamma K^{1/2} A K^{1/2}\bigr)$. The Schur complement of the lower right block λ gives $\det(\lambda - \Xi) = p(\lambda)$ as well, for the self-adjoint matrix

$$\Xi = \begin{pmatrix} \gamma K^{1/2} A K^{1/2} + b^2 X_+ - 1 & bX_+^{1/2} \\ bX_+^{1/2} & 0 \end{pmatrix},$$

which is affine in A. Since $\det(\lambda - YXY) = \det(\lambda - XY^2)$, the polynomials agree and YXY and Ξ have the same eigenvalues; hence $\operatorname{Tr}|YXY| = \|\Xi\|_1 \ge \operatorname{Tr}(T\Xi)$ for every self-adjoint T with $\|T\| \le 1$. Take

$$T = \begin{pmatrix} -\Lambda & G \\ G & \Lambda \end{pmatrix}, \qquad \Lambda = \frac{1-X_+}{1+X_+}, \quad G = \frac{2X_+^{1/2}}{1+X_+},$$

so that $T^2 = 1$ (Λ and G commute and $\Lambda^2 + G^2 = 1$). With $\operatorname{Tr}(Y|X|Y) = \operatorname{Tr}(|X|Y^2) = b^2 \operatorname{Tr} X_+ + \gamma \operatorname{Tr}(X_+ A) + mb^2 + \gamma \operatorname{Tr}(1-A)$ and $K^{1/2}\Lambda K^{1/2} = 1 - X_+$, the terms containing A cancel in $\operatorname{Tr}(|X|Y^2) - \operatorname{Tr}(T\Xi)$, and

$$\operatorname{Tr}(Y|X|Y) - \operatorname{Tr}|YXY| \le \sum_{i=1}^{m} \Bigl(b^2 x_i + 1 + \frac{1-x_i}{1+x_i}(b^2x_i - 1) - \frac{4bx_i}{1+x_i} \Bigr) = (1-b)^2 \sum_{i=1}^{m} \frac{2x_i}{1+x_i}.$$

For diagonal X_+ both operators are direct sums of 2×2 blocks, on which $X = \operatorname{diag}(x_i, -1)$, and Proposition 4.2 gives the value $(1-b)^2\, 2x_i/(1+x_i)$ at the stated angle (there $\gamma = \sin^2\theta_i$). □

Lemma 4.5. *Let $X = X^*$ be invertible, P the projection onto the positive part of X, $S = 2P - 1$, and $Y \ge 0$. Then (a) if $|X| \le |X'|$, $X' = S|X'|$ and $|X'|$ commutes with P, then $\mathcal{D}(X,Y) \le \mathcal{D}(X',Y)$; (b) for fixed X, $\mathcal{D}$ is a convex function of Y^2.*

Proof. With $A = YX_+Y$ and $B = YX_-Y$, $\mathcal{D} = \operatorname{Tr} A + \operatorname{Tr} B - \|A - B\|_1$. (a) $A = Y|X|^{1/2}P|X|^{1/2}Y$ is non-decreasing in $|X|$ among operators commuting with P, and so is B; by the triangle inequality $\|A + D - B\|_1 \le \|A - B\|_1 + \operatorname{Tr} D$ for $D \ge 0$, so $\mathcal{D}$ does not decrease when A or B increases. (b) As $\|H\|_1 = \max_{-1\le T\le 1} \operatorname{Tr}(TH)$ for self-adjoint H, with $T = 1 - 2E$ we get $\mathcal{D} = 2\min_{0\le E\le 1} \operatorname{Tr}\big(EA + (1-E)B\big)$. For invertible Y, $Z = |X|^{1/2}Y$ is invertible and $E \mapsto F = ZEZ^*$ maps $\{0 \le E \le 1\}$ onto $\{0 \le F \le G\}$, $G = ZZ^* = |X|^{1/2}Y^2|X|^{1/2}$, with $\operatorname{Tr}(EA) = \operatorname{Tr}(PF)$ and $\operatorname{Tr}((1-E)B) = \operatorname{Tr}((1-P)(G-F))$. So $\mathcal{D} = 2\min\{\operatorname{Tr}(PF + (1-P)(G-F)) : 0 \le F \le G\}$. For two positive G_0, G_1 choose minimisers F_0, F_1. Their convex combination is feasible for the corresponding convex combination of G_0, G_1, and its objective is the same convex combination of the two minimal objectives. Thus the minimum at the combined G is at most that value, proving convexity in G, which is affine in Y^2. Continuity gives the same conclusion at singular Y. □

Proof of Theorem 4.3. Write $a = \ell$, $b = \ell'$; it suffices to treat the complex algebra, since the maximisers are real. If $\delta' = 0$, then $Y = I$ and the defect vanishes, as does the stated maximum. Otherwise $b < 1$. Let $\delta < 1$, so that X is invertible. Let P be its positive spectral projection and $S = 2P - I$; one has $\mathcal{D}(-X, Y) = \mathcal{D}(X, Y)$. Since $|X|$ commutes with P, it has a unit eigenvector e for its eigenvalue a in one sector and in $\operatorname{ran} P$ or in $\ker P$; replacing X by $-X$ if necessary, $e \in \operatorname{ran} P$. Then $|X| \le |X'| := 1 - (1-a)ee^*$, and $X' = S|X'| = S - (1-a)ee^*$ has contrast δ; by Lemma 4.5(a), $\mathcal{D}(X, Y) \le \mathcal{D}(X', Y)$. By Lemma 4.5(b), a convex function of the eigenvalues of Y^2 on the faces $\{y_i = 1,\ y_j = b^2\}$ of $[b^2, 1]^K$ is maximal at a vertex, so $\mathcal{D}(X', Y) \le \mathcal{D}(X', Y')$ with $Y' = b + (1-b)Q$, $Q \ne 0, 1$ a projection in $\mathcal{M}$. Apply Lemma 2.3 to P and Q in every sector. In a sector without e, $X' = S$ is a direct sum over the subspaces, a one-dimensional summand contributes 0, and a two-dimensional one at most $(1-b)^2$ by Proposition 4.2 ($|X| = 1$ there): together at most $\lfloor k_c/2 \rfloor (1-b)^2$. In the sector of e, Lemma 4.5(a) with $|X'| \le 1$ gives the same bound for $\mathcal{D}$ restricted to it, and, if its size k_c is odd or it contains fewer than $k_c/2$ two-dimensional subspaces, at most $\lfloor k_c/2 \rfloor (1-b)^2$, resp. $(k_c/2 - 1)(1-b)^2$. Otherwise its $m = k_c/2$ two-dimensional subspaces fill it, with bases f_j, f_j', $f_j \in \operatorname{ran} P$. Choose the phases of f_j' so that the off-diagonal entries of Q in each principal-angle block are real and nonnegative. This change of basis commutes with X', which is $-I$ on $\ker P$. In the basis $(f_1, \dots, f_m, f_1', \dots, f_m')$ we are in the setting of Proposition 4.4 with $X_+ = 1 - (1-a)ee^*$, whose eigenvalues are $a, 1, \dots, 1$: the sector contributes at most $(1-b)^2\big(\frac{2a}{1+a} + m - 1\big) = (k_c/2 - \delta)(1-b)^2$. Summing over the sectors, $\mathcal{D} \le (\nu - \delta)(1-b)^2$ if every k_c is even and $\mathcal{D} \le \nu(1-b)^2$ otherwise. Equality: if every k_c is even, $X = \operatorname{diag}(a, -1) \oplus \operatorname{diag}(1, -1)^{\oplus(\nu-1)}$ and Y with eigenvalues $1, b$ in every block at the angle of Proposition 4.2; if some k_c is odd, $X = \operatorname{diag}(1, -1)$ on ν blocks, the eigenvalue a on a one-dimensional block of an odd sector, 1 on the other one-dimensional blocks, and Y as before ($Y = 1$ on the one-dimensional blocks). □

5 Conclusion and scope

The two extremal formulas use the same spectral constraints but have different parity mechanisms. For the Gardner quotient, a scalar summand of positive weight dilutes the two-dimensional ratio; the resulting odd-dimensional optimum has an explicit radical form. For the trace-norm defect, an unused scalar direction can hold the smallest modulus eigenvalue without lowering the contribution of the active two-dimensional blocks. When every sector is even, the matrix linearisation proves the exact loss even if that eigenvector is not supported in one principal-angle block.

The arguments are analytic. Numerical consistency checks made during preparation are not proofs of the global maxima and no historical research dataset has been regenerated for this focused text. Self-adjointness of X is required for the defect result; the normalised Gardner theorem requires

$W > 0$. Zero contrasts and singular modulus endpoints are treated only as specified in their statements.

Disclosure of AI assistance

OpenAI Codex, a language-model assistant, contributed to restructuring and drafting the scientific text, exploring and developing the explicit odd-dimensional optimisation, and developing and executing numerical consistency checks. Its involvement was not limited to language editing. The sharp bounds here are supported by the analytic proofs stated in the manuscript; floating-point checks are not used as proofs. Responsibility for assessing correctness, originality, attribution and the final submission rests with the human authors.

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